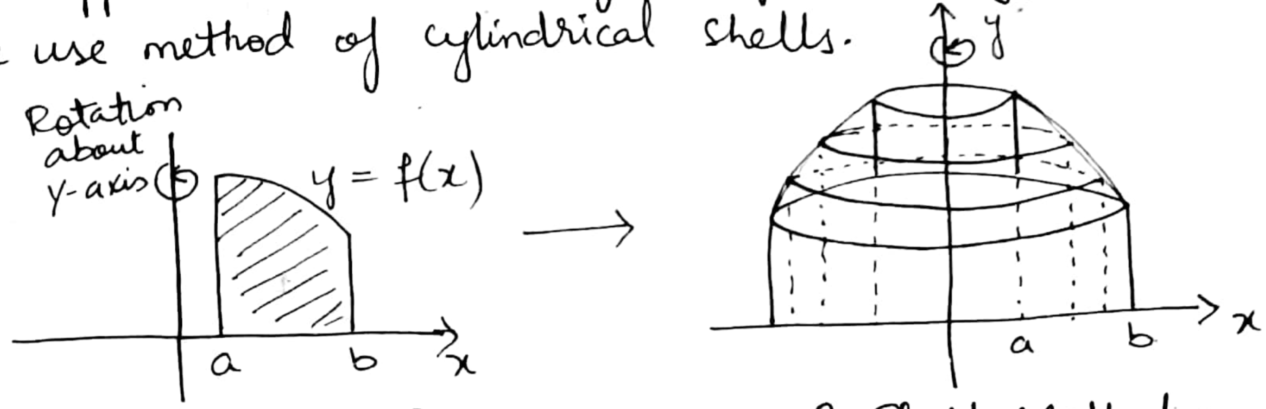
Topic 6.2

Volumes Using Cylindrical Shells

- **Cylindrical Shells**

Problem: Let "f" be cont. and non-negative on $[a, b]$, and let R be the region that is bounded above by $y = f(x)$, below by x-axis, and on the sides by the lines $x = a$, and $x = b$. Find the volume "V" of the solid of revolution "S" that is generated by revolving by the region R about y-axis.

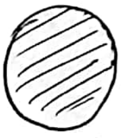
Note: Sometime method of "Disks" and "Washer" not applicable or resulting integral is difficult, so we use method of cylindrical shells.



Difference Between Disk, Washer & Shell Method:

	Disk Method	Washer Method	Shell Method
Shape Used	Disk (Solid circle)	Washer (ring-shaped)	Shell (Hollow Cylindrical Layer)
Orientation	Perpendicular to the axis of rotation	Perpendicular to the axis of rotation	Parallel to the axis of rotation.
When to Use	Solid region (No Hole)	Region with a hole (between two curves)	Axis of rotation outside the region.
Volume Formula	$V = \pi\int_a^b [f(x)]^2\,dx$ or $V = \pi\int_c^d [f(y)]^2\,dy$	$V = \pi\int_a^b ([f(x)]^2 - [g(x)]^2)\,dx$ or $V = \pi\int_c^d [[f(y)]^2 - [g(y)]^2]\,dy$	$V = 2\pi\int_a^b (\text{radius})(\text{height})\,dx$ or $V = 2\pi\int_c^d (\text{radius})(\text{height})\,dy$.

Disk / Washer Method: The volume is calculated by slicing the solid into cross-sections (disks or washers) that are perpendicular to the axis of rotation. Each cross-section is then integrated to find the total volume.

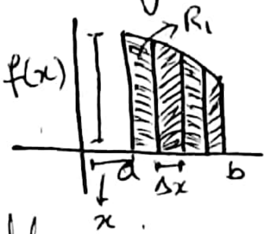
$V=\int_a^b \pi[f(x)]^2\,dx$ ← 

(Disk)

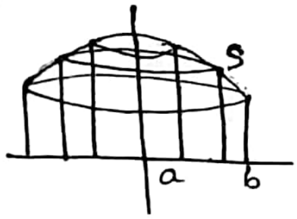
 → $V=\int_a^b \pi\left[(f(x))^2-(g(x))^2\right]dx$

(Washer).

Shell Method: The volume is calculated by revolving strips (thin rectangles) that are parallel to the axis of rotation. These strips create thin cylindrical shells, and the volume is obtained by summing the volumes of these shells.

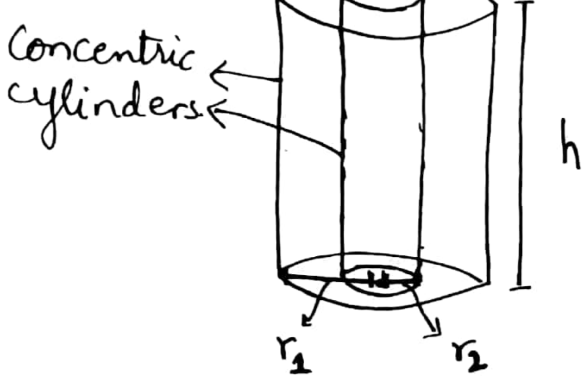


⟶



Cylindrical Shell: is a solid enclosed by two concentric right circular cylinder. The volume "V" of cylindrical shell with inner radius "r_2" and outer radius "r_1" and height "h".

- Concentric Right Circular Cylinder refers to two or more cylinders that share the same center and axis but may have different radii or heights. The term "Concentric" means that they are aligned along the same central point or axis.



gion: $V = [\text{Area of cross-section}] \times \text{Height}$

$$V = [\pi r_1^2 - \pi r_2^2] \cdot h$$

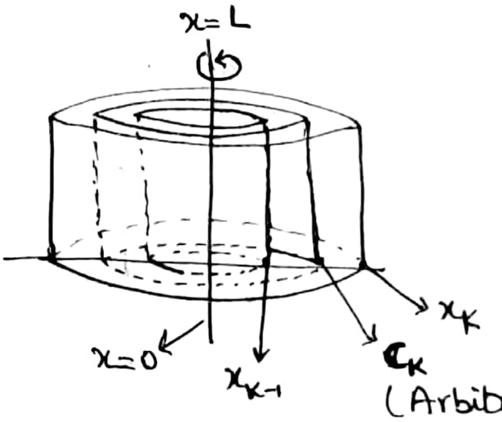
$$V = \pi [r_1^2 - r_2^2] \cdot h$$

$$V = \pi (r_1 + r_2)(r_1 - r_2) \cdot h$$

$$V = 2\pi \cdot \frac{1}{2} (r_1 + r_2)(r_1 - r_2) \cdot h$$

$$V = 2\pi \left[\frac{1}{2}(r_1 + r_2) \right] \cdot h \cdot (r_1 - r_2).$$

$$\boxed{V = 2\pi \cdot \left(\begin{array}{c}\text{Average}\\ \text{Radius}\end{array}\right) \cdot (\text{Height}) \cdot (\text{Thickness})}$$



- Make a cut at c_k, the midpoint of the section (Average)
- Thickness is Δx.
- Height is $f(c_k)$.

$$V = 2\pi \cdot \underbrace{c_k}_{(c_k - L)} \cdot f(c_k) \cdot \Delta x \rightarrow \text{Formula for one individual shell of } c_k$$

$$V = \sum_{k=1}^{n} 2\pi \cdot (\underbrace{c_k}_{(c_k - L)}) \cdot f(c_k) \cdot \Delta x \rightarrow \text{Approximation of all shells}$$

$$V = \lim_{n \to \infty} \sum_{k=1}^{n} 2\pi \cdot (\underbrace{c_k}_{(c_k - L)}) \cdot f(c_k) \cdot \Delta x \rightarrow \text{Exact volume.}$$

$$V = \int_a^b 2\pi (x)(f(x))\, dx.$$

or

Book Formula ← $$V = \int_a^b 2\pi (x - L) f(x)\, dx.$$

↓ About y-axis.

$$V = \int_c^d 2\pi y\, g(y)\, dy.$$

Book Formula

or

$$V = \int_c^d 2\pi (y - L) g(y)\, dy$$

↓ About x-axis.

Definition: The volume of the solid generated by revolving the region between the x-axis and the graph of a continuous function $y = f(x) \geq 0$, $L \leq a \leq x \leq b$ about a vertical line $x = L$ is:

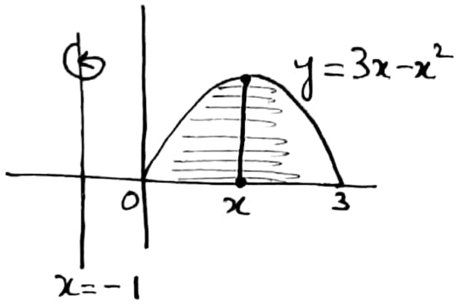
$$V = \int_a^b 2\pi \underbrace{\begin{pmatrix} \text{shell} \\ \text{radius} \end{pmatrix}}_{\substack{\text{Distance from} \\ \text{axis of rotation to} \\ \text{a typical rectangle}}} \underbrace{\begin{pmatrix} \text{shell} \\ \text{height} \end{pmatrix}}_{\substack{\text{Height of a} \\ \text{typical} \\ \text{rectangle}}} \, dx$$

For a vertical axis of rotation; When the axis is horizontal, use dy.

Example: Rotate the region between the curve $y = f(x) = 3x - x^2$ and the x-axis about the $x = -1$. Find the volume.

Sol:

$$V = \int_0^3 2\pi \left(x - (-1)\right)\left(3x - x^2\right) dx$$



Axis of Rotation

$$V = \int_0^3 2\pi (x+1)\,(3x - x^2)\,dx$$

$$V = \int_0^3 2\pi\left(3x^2 - x^3 + 3x - x^2\right) dx$$

$$V = \int_0^3 2\pi\left(2x^2 + 3x - x^3\right) dx$$

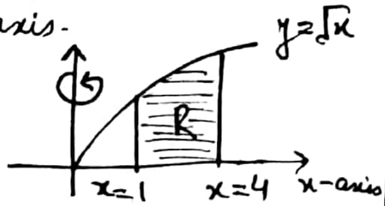
$$V = 2\pi \int_0^3 \left(2x^2 + 3x - x^3\right) dx$$

$$V = 2\pi\left[\frac{2x^3}{3} + \frac{3x^2}{2} - \frac{1}{4}x^4\right]_0^3$$

$$\boxed{V = \frac{45\pi}{2}}$$ Ans

$\longleftrightarrow$

Find the volume of the region inclosed by $y=\sqrt{x}$, $x=1$ and $x=4$ when revolved about y-axis. ⑤



Sol:

$$V=\int_1^4 2\pi(x-0)(\sqrt{x})\,dx$$

$$V=\int_1^4 2\pi(x)(\sqrt{x})\,dx = 2\pi\int_1^4 x^{3/2}\,dx$$

$$V=2\pi\left.\frac{x^{5/2}}{5/2}\right|_1^4 = \frac{4}{5}\pi\left[x^{5/2}\right]_1^4 = \frac{4\pi}{5}\left[4^{5/2}-1^{5/2}\right]$$

$$\boxed{V=\frac{124\pi}{5}}$$ Ans

Q Revolve the region bound by $y=x$ and $y=x^2$ around the y-axis. Find the volume.

Sol: Intersect point:

$$x=x^2$$
$$0=x^2-x$$
$$x(x-1)=0$$
$$\boxed{x=0},\ \boxed{x=1}$$

Formula

$$V=\int_a^b 2\pi(x-L)f(x)\,dx$$

Function is given in terms of x and the region is created around y-axis So we use Shell Method

$$V=\int_0^1 2\pi(x-0)[x-x^2]\,dx$$

$0 \quad \frac{1}{2} \quad 1$

Put $x=\frac{1}{2}$

$x=\frac{1}{2}$, $x^2=\left(\frac{1}{2}\right)^2=\frac{1}{4}$

Top Bottom

$$V=\int_0^1 2\pi(x)(x-x^2)\,dx$$

$$V=\int_0^1 2\pi(x^2-x^3)\,dx$$

$$V=2\pi\left[\frac{x^3}{3}-\frac{x^4}{4}\right]_0^1 = 2\pi\left[\left(\frac{1}{3}-\frac{1}{4}\right)-0\right] = 2\pi\left(\frac{1}{12}\right)$$

$$\boxed{V=\frac{\pi}{6}}$$ Ans

Q Revolve the region bound by $x = -y^2 + 6y$ and $x = 0$ around x-axis.

Sol: $V = \int_c^d 2\pi (y - L)\, g(y)\, dy$ Formula.

Intersect Point: $-y^2 + 6y = 0$

$-y(y-6) = 0$

$\boxed{y=0}, \boxed{y=6}$

Put $x = 1$

0 6

$x = -1^2 + 6 = 5$ (Right)

$x = 0$ (Left)

$$V = \int_0^6 2\pi (y - 0)(-y^2 + 6y)\, dy$$

$$V = \int_0^6 2\pi (y)(-y^2 + 6y)\, dy$$

$$V = \int_0^6 2\pi (-y^3 + 6y^2)\, dy = 2\pi \int_0^6 (-y^3 + 6y^2)\, dy.$$

$$V = 2\pi \left[-\frac{y^4}{4} + \frac{6y^3}{2} \right]_0^6 = 2\pi \left[\left(-\frac{(6)^4}{4} + 2(6)^3 \right) - 0 \right]$$

$$\boxed{V = 216\pi}$$ Ans